Fire Extinguishing System – General Data

1. General

The fire extinguishing system is comprised of two independent systems, one for each engine. Each system has a warning light/push button located in the flight compartment; which when pushed, electrically discharges a fire extinguisher bottle. The extinguishing agent from the bottle is directed by a discharge nozzle into the appropriate engine accessory compartment.

2. Description

Each engine fire extinguishing system consists of a fire extinguisher bottle with discharge pipes and indicator discs, and a fire extinguisher Warning Light/Push Button. Figure 1 shows component locations and system details.

A. Fire Extinguisher Bottles

A fire extinguisher bottle containing HALON 1301, is installed in each engine nacelle behind the firewall. A main discharge pipe extends from a valve on the bottom of the bottle through the firewall to the engine accessory compartment. Two additional pipes, one connected to the bottle and the other to the main discharge pipe, terminate at indicator discs on the side of the nacelle.

B. Fire Extinguisher Bottle Indicating Discs

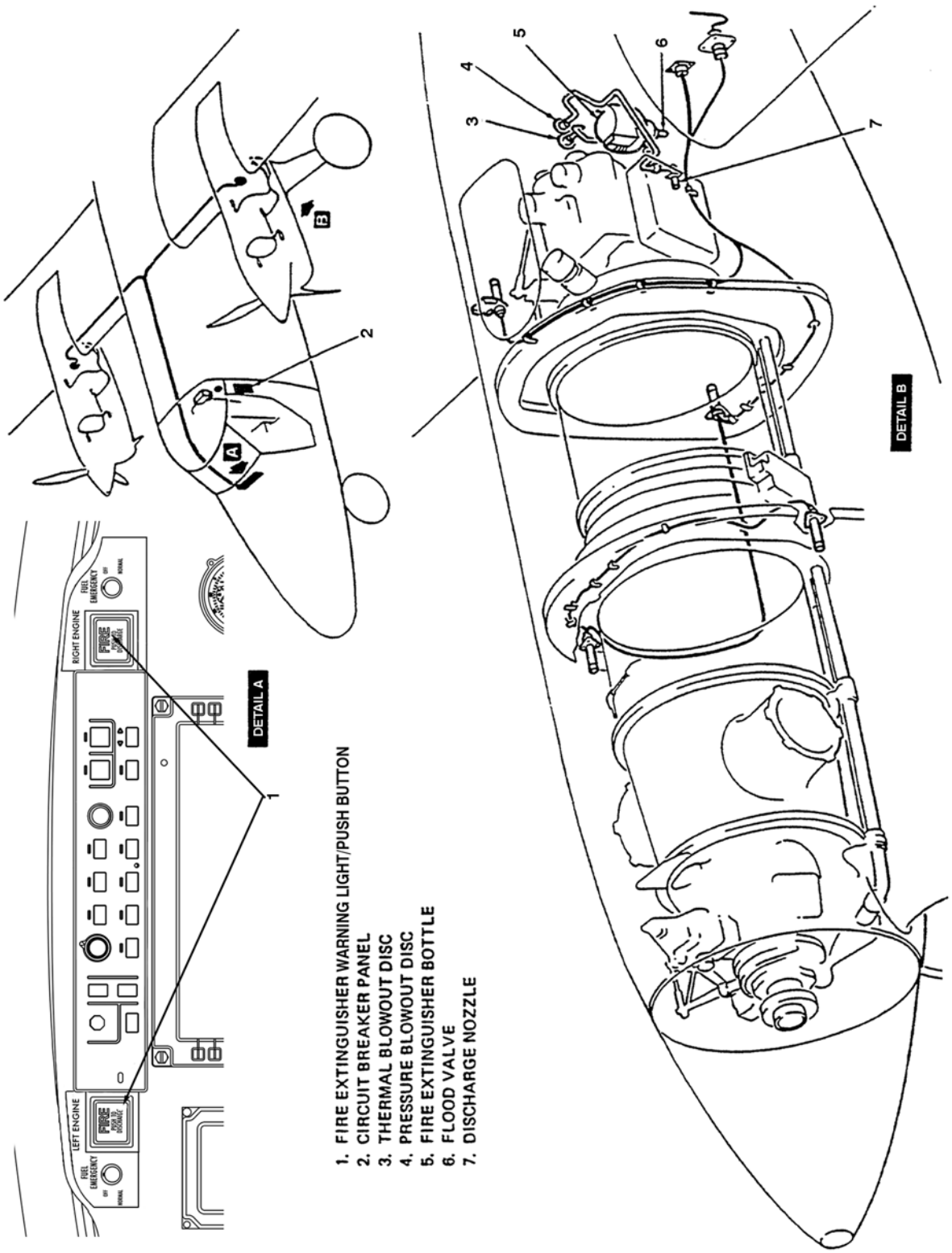
Two fire extinguisher bottle indicating discs are provided for each bottle, one yellow, and the other red; they are located on the inboard side of each engine nacelle.

C. Fire Extinguisher Warning Light/Push Buttons

A fire extinguisher warning light/push button, identified FIRE PUSH TO DISCHARGE, is provided for each engine system.

3. Operation

The warning light/push button operated switch of each system is supplied with 28 Volt DC from its respective FIRE EXTINGUISHER L or FIRE EXTINGUISHER R 5-ampere circuit breaker. When the FIRE push button is pushed, the switch is operated, and the circuit is completed to the appropriate bottle valve which is opened by the action of a squib assembly. The fire extinguishing agent is then expelled through the valve and main discharge pipe into the engine accessory compartment. The yellow indicator disc, which is backed up by a piston assembly in the pipe connected to the main discharge pipe, is punched out when the extinguisher operates. When the fire bottle is discharged due to thermal expansion, the red disc is blown out.



Fire Extinguishing System – Component Locations
Figure 1

4. Adjustment/Test

A. Function Test Fire Extinguishing System

WARNING DO NOT USE A VOLT-OHMMETER, BATTERY POWERED CONTINUITY LIGHT, OR ANY SIMILAR DEVICE IN AN ATTEMPT TO TEST THE SQUIB ASSEMBLY. USE OF A MULTIMETER, WHICH PASSES MORE THAN 25 MILLIAMP, MAY DETONATE THE SQUIB ASSEMBLY.

Equipment Required:

- Multimeter – that electrically limits test current to 25 mA, maximum
- Pressure test rig (refer to 21-40-00, Heating System – General Data)
- Test lamp-28 volt

Note Before carrying out the following tests ensure fire extinguisher system is completely and correctly installed.

1) Safety Precautions

a) All personnel shall stand clear of the fire extinguisher discharge nozzles at all times when FIREX system tests are being carried out.

b) Only a multimeter that electrically limits test current to 25 milliamps maximum shall be used to check firing cartridge continuity.

c) The use of a volt-ohmmeter, battery powered continuity light or any similar device to test the firing cartridge is prohibited as these devices may detonate the cartridge.

d) If a FIREX bottle is accidently discharged in an aircraft, the aircraft shall be well ventilated before personnel are allowed to enter.

e) Frostbite or low-temperature burns will result if the fire extinguishing agent (HALON 1301) is allowed to contact the skin.

2) Pressure Test

a) Remove split ring and the yellow indicator disc from discharge indicator on the inboard side of the right engine nacelle.

b) Connect pressure hose from test rig to end of discharge nozzle just forward of firewall in right engine nacelle.

c) Adjust test rig to supply 25 psi, check that piston at discharge indicator moves outwards as pressure applied, and reseats when pressure is reduced.

d) Disconnect test rig, replace yellow indicator disc and split ring in discharge indicator.

e) Ensure split ring is fully seated in its retaining groove.

f) Repeat Para 4.A.2).a) through Para 4.A.2).e) for left engine system.

3) System Circuit Test

a) Ensure that all aircraft electrical power is switched OFF and that all circuit breakers are disengaged (refer to 24-00-00, Electrical Power – General Data).

b) Disconnect the positive electrical wire from the insulated post on the LH and RH FIREX bottle firing cartridge. Leave the ground wire connected to the bottles. Connect a 28 Volt test lamp between the disconnected wiring at each FIREX bottle and airframe ground.

c) Engage electrical power circuit breakers. Select POWER SOURCE switch to BATTERY, and MASTER switch to ON (Pre Mod 6/2314); or select BATTERY switch to ON (Post Mod 6/2314). Engage FIRE EXTINGUISHER L and FIRE EXTINGUISHER R circuit breakers.

d) Push the RIGHT ENGINE–FIRE warning push button located on the emergency panel and check that RH test lamp illuminates. Release and check that test lamp is extinguished.

e) Push the LEFT ENGINE–FIRE warning push button located on the emergency panel and check that LH test lamp illuminates. Release and check that test lamp is extinguished.

f) Switch off aircraft DC power, disengage FIRE EXTINGUISHER L and R circuit breakers and remove test lamps from FIREX bottle wiring.

g) Reconnect the electrical wire to the FIREX bottles. Ensure that ground wire (identified by suffix letter N in wire code) is connected to ground terminal and positive wire is connected to actuate terminal.

h) Engage FIRE EXTINGUISHER L and FIRE EXTINGUISHER R circuit breakers.

4) Continuity Test

WARNING **DO NOT USE A VOLT-OHMMETER, BATTERY POWERED CONTINUITY LIGHT, OR ANY SIMILAR DEVICE IN AN ATTEMPT TO TEST THE SQUIB ASSEMBLY. USE OF A MULTIMETER, WHICH PASSES MORE THAN 25 MILLIAMP, MAY DETONATE THE SQUIB ASSEMBLY.**

a) Immediately prior to using the multimeter, short out the test leads, and check that the meter current is no more than 10 – 25 milliamps.

b) Connect leads from the multimeter to the firing cartridge terminal studs on the right engine FIREX bottle, and check that meter indicates continuity.

c) Disconnect test box and reconnect FIREX bottle wiring to firing cartridge terminals.

d) Ensure that ground wire (identified by suffix letter N in wire code) is connected to ground terminal and positive wire is connected to actuate terminal.

e) Repeat Para 4.A.4).b) through Para 4.A.4).d) for left engine FIREX bottle.

f) Check pressure gauges on FIREX bottles each indicated fully charged condition in accordance with the manufacturers' data plate (450 psi at 70°F).

g) Ensure firing cartridge has been wirelocked to its FIREX bottle.

h) At completion of test, position the rubber nipple on the actuate wire and secure with a plastic tie.